Practice workbook · Factorising and solving a cubic

Dividing a cubic by a known linear factor and factorising the quadratic quotient, to write f(x) as a product of factors or to solve f(x) = 0.

22 question parts: 1 worked-example retry and 21 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-10

1. Question 3 · part b · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

3 Given that $(x - 4)$ is a factor of $px^3 - 31x^2 + 25x + 12$ where p is a constant,

(a) show that $p = 6$

(2)

(b) Solve the equation $6x^3 - 31x^2 + 25x + 12 = 0$

Show clear algebraic working.

(4)

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1. Question 3 · part b · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

1. Question 3 · part b · 4 marks · Working and answer

Working and answer

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2. Question 9 · part b.ii · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

9

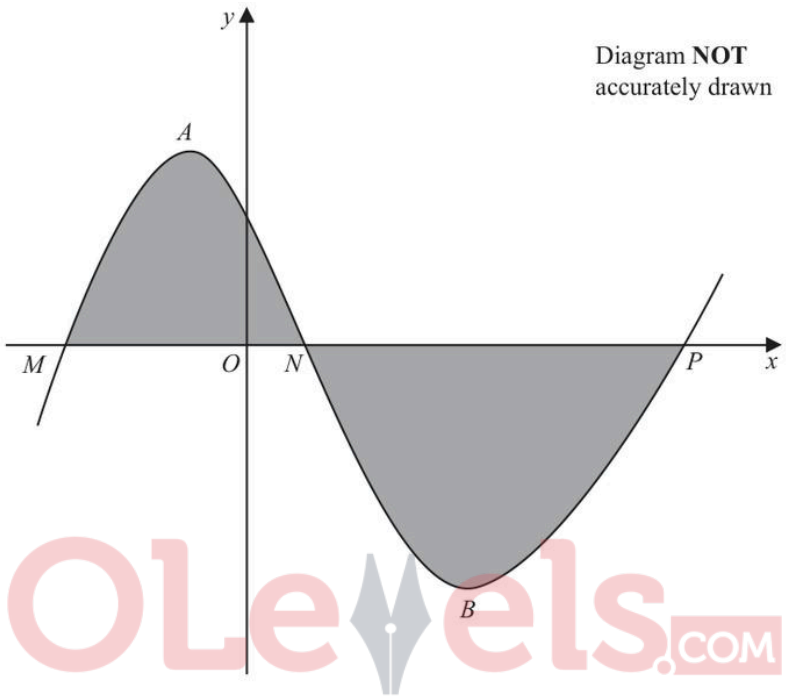


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^3 + ax^2 + bx + c$$

The curve C has a maximum at the point A with coordinates $\left(-\frac{1}{3}, \frac{100}{27}\right)$ and a minimum at the point B with coordinates $(2, -9)$

Given that a, b and c are integers

(a) show that $a = -5, b = -4$ and $c = 3$

(5)

(b) (i) Show that $(x+1)$ is a factor of $f(x)$

(1)

(ii) Hence, or otherwise, use algebra to factorise $f(x)$ completely.

(3)

The curve C crosses the x-axis at the points M, N and P
The finite regions shown shaded in Figure 4 are bounded by the curve C and parts of the x-axis from M to N and from N to P

(c) Use algebraic integration to determine the total area of the shaded regions. Give your answer as an exact fraction.

(4)

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2. Question 9 · part b.ii · 3 marks · Working and answer

Working and answer

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3. Question 2 · part c · 3 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

2 $f(x) = x^3 + px + q$ where p and q are constants.

The remainder when $f(x)$ is divided by $(x - 1)$ is -12

The remainder when $f(x)$ is divided by $(x - 4)$ is 30

(a) Find the value of p and the value of q.

(6)

Using your values of p and q

(b) show that $f(3) = 0$

(1)

(c) Express $f(x)$ as a product of linear factors.

(3)

(d) Hence solve the equation $f(x) = 0$

(1)

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3. Question 2 · part c · 3 marks · Working and answer

Working and answer

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4. Question 4 · part b · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

4 $f(x) = 2x^3 + ax^2 + bx + 18$ where a and b are constants

When $f'(x)$ is divided by $(x - 2)$ the remainder is 5

Given that $(x - 2)$ is a factor of $f(x)$

(a) find the value of a and the value of b.

(6)

(b) Express $f(x)$ as a product of linear factors.

(3)

(c) Hence use algebra to solve the equation $f(x) = 0$

(2)

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4. Question 4 · part b · 3 marks · Working and answer

Working and answer

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5. Question 5 · part b · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

5 $f(x) = 2x^3 + ax^2 - 14x + b$ where a and b are constants.

When $f(x)$ is divided by $(x - 4)$ the remainder is 39

Given that $(x - 1)$ is a factor of $f(x)$

(a) show that $a = -3$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(4)

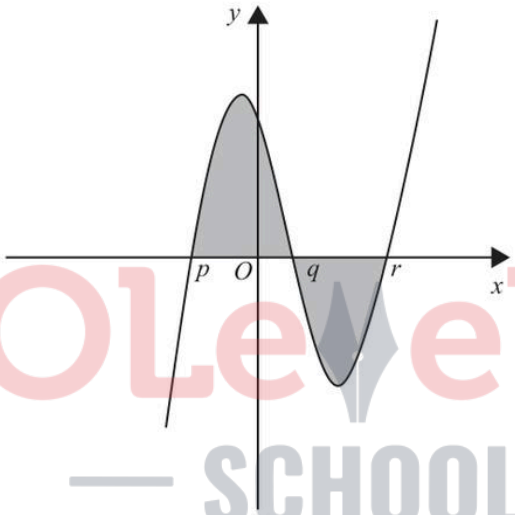


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Figure 3

Figure 3 shows part of the curve C with equation $y = f(x)$

Given that C crosses the x-axis at the points with coordinates $(p, 0)$, $(q, 0)$ and $(r, 0)$

(c) write down the value of p, the value of q and the value of r

(3)

The region shown shaded in Figure 3 is bounded by the curve and the x-axis.

(d) Use algebraic integration to find the exact area of the shaded region.

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5. Question 5 · part b · 4 marks · Working and answer

Working and answer

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6. Question 4 · part b · 3 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

4 $f(x) = px^3 + qx^2 - 37x - 12q$ where p and q are constants.

When $f'(x)$ is divided by $(x + 2)$ the remainder is -33

Given that $(x + 5)$ is a factor of $f(x)$

(a) (i) show that $p = 2$

(ii) find the value of q

(6)

(b) Hence, use algebra to factorise $f(x)$ completely.

(3)

(c) Hence solve the equation $f(x) = 0$

(2)

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6. Question 4 · part b · 3 marks · Working and answer

Working and answer

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7. Question 6 · part d · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part d only.

6 $f(x) = x^3 + (p + 1)x^2 - 10x + q$

where p and q are integers.

Given that $(x - 3)$ is a factor of $f(x)$

(a) show that $9p + q + 6 = 0$

(3)

Given that $(x + p)$, where $p > 0$, is also a factor of $f(x)$

(b) show that $p^2 + 10p + q = 0$

(3)

(c) Hence find the value of p and the value of q.

(5)

(d) Using your values of p and q, factorise $f(x)$ completely.

(2)

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7. Question 6 · part d · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part d only.

Question 6 continued

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7. Question 6 · part d · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part d only.

Question 6 continued

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7. Question 6 · part d · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part d only.



Question 6 continued

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(Total for Question 6 is 13 marks)

7. Question 6 · part d · 2 marks · Working and answer

Working and answer

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8. Question 9 · part a · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part a only.

9 $f(x) = 3x^4 + 4x^3 - 36x^2 + 64$

Given that $f(x)$ can be written in the form $(x-2)^2(ax^2 + bx + c)$

(a) find the value of a, the value of b and the value of c

(4)

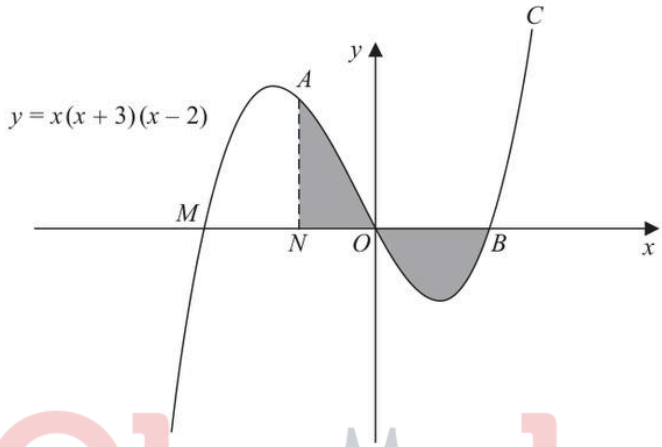


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Figure 3

Figure 3 shows a sketch of part of the curve C with equation $y = x(x+3)(x-2)$

The curve C crosses the x-axis at the point M, the origin and the point B.
The point N lies on the x-axis between M and O.
The point A lies on C such that AN is parallel to the y-axis.

The area of the shaded region bounded by the curve and OB is numerically equal to the area of the shaded region bounded by the curve, ON and NA.

Given that the coordinates of N are $(n, 0)$,

(b) use algebraic integration to show that n satisfies the equation

$$(x-2)^2(3x^2 + 16x + 16) = 0$$

(7)

(c) Hence find the exact coordinates of A.

(5)

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8. Question 9 · part a · 4 marks · Working and answer

Working and answer

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9. Question 5 · part b · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

5 $f(x) = 2ax^3 + x^2 - bx + 3a$ where a and b are integers.

Given that $(x + 2)$ and $(x - 1)$ are both factors of $f(x)$

(a) show that $a = 2$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(2)

Hence, given that $h(y) = 2^{(3y+2)} + 2^{2y} - 11(2^y) + 6$

(c) solve the equation $h(y) = 0$

Where appropriate give your answers to 3 decimal places.

(5)

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9. Question 5 · part b · 2 marks · Working and answer

Working and answer

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10. Question 5 · part c · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

5 $f(x) = 2x^3 + ax^2 - 14x + b$ where a and b are constants.

When $f(x)$ is divided by $(x - 4)$ the remainder is 39

Given that $(x - 1)$ is a factor of $f(x)$

(a) show that $a = -3$ and find the value of b

(5)

(b) Hence factorise $f(x)$ completely.

(4)

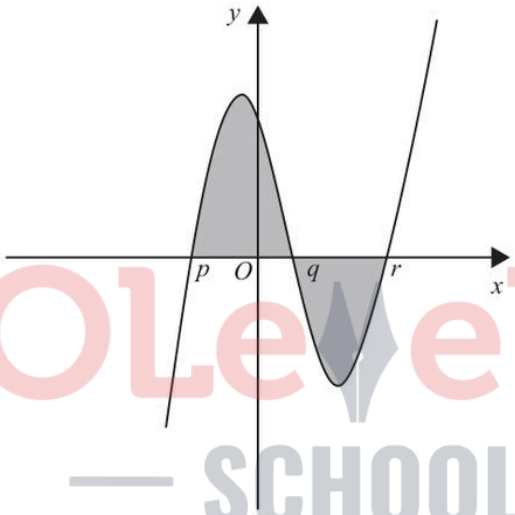


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = f(x)$

Given that C crosses the x-axis at the points with coordinates $(p, 0)$, $(q, 0)$ and $(r, 0)$

(c) write down the value of p, the value of q and the value of r

(3)

The region shown shaded in Figure 3 is bounded by the curve and the x-axis.

(d) Use algebraic integration to find the exact area of the shaded region.

(4)

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10. Question 5 · part c · 3 marks · Working and answer

Working and answer

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11. Question 5 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

5 $f(x) = ax^3 + 5bx^2 + 8ax - 4b$ where a and b are integers

Given that

$(x + 2)$ is a factor of $f(x)$

and that

when $f(x)$ is divided by $(x + 3)$ the remainder is 21

(a) show that $a = 2$ and find the value of b

(5)

(b) Use algebra to solve the equation $f(x) = 0$

(4)

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11. Question 5 · part b · 4 marks

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Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 9 marks)

11. Question 5 · part b · 4 marks · Working and answer

Working and answer

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12. Question 4 · part b · 3 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

4 $f(x) = x^3 + px^2 + qx + 6$ where p and q are constants.

Given that $(x - 1)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 8

(a) (i) show that $p = -2$

(ii) find the value of q

(6)

(b) Hence, solve the equation $f(x) = 0$

(3)

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12. Question 4 · part b · 3 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 9 marks)

12. Question 4 · part b · 3 marks · Working and answer

Working and answer

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13. Question 2 · part d · 1 mark

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

2 $f(x) = x^3 + px + q$ where p and q are constants.

The remainder when $f(x)$ is divided by $(x - 1)$ is -12

The remainder when $f(x)$ is divided by $(x - 4)$ is 30

(a) Find the value of p and the value of q.

(6)

Using your values of p and q

(b) show that $f(3) = 0$

(1)

(c) Express $f(x)$ as a product of linear factors.

(3)

(d) Hence solve the equation $f(x) = 0$

(1)

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13. Question 2 · part d · 1 mark

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 2 continued

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(Total for Question 2 is 11 marks)

13. Question 2 · part d · 1 mark · Working and answer

Working and answer

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14. Question 10 · part b · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

10 $f(x) = 32x^3 - 33x + 1$

(a) Show that $f(1) = 0$

(1)

(b) Hence using an algebraic method solve $f(x) = 0$

(4)

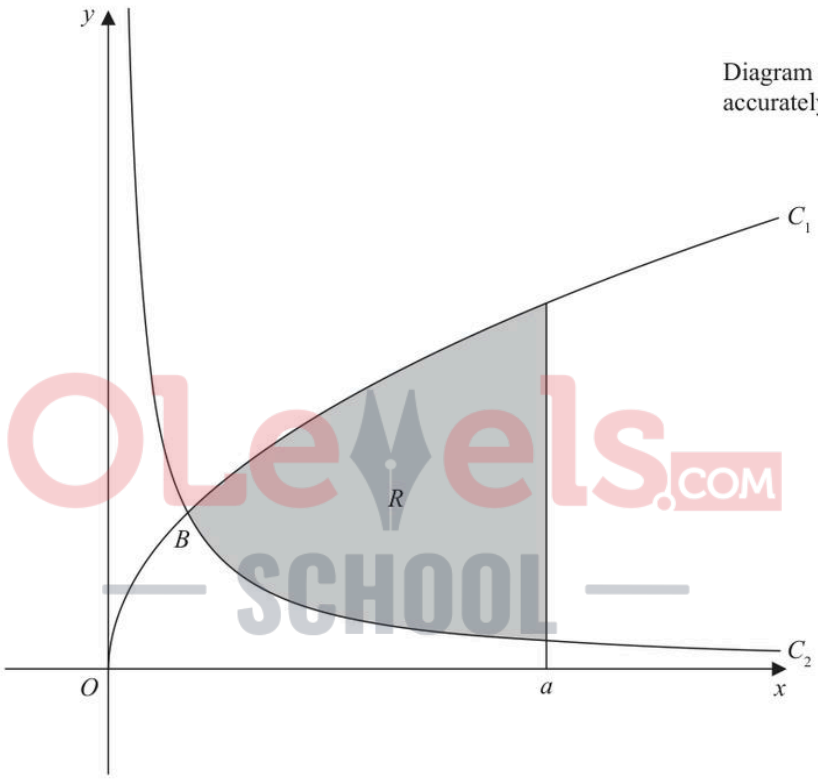


Diagram **NOT** accurately drawn

Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve C_1 with equation $y = \sqrt{x}$, by the curve C_2 with equation $y = \frac{1}{8x}$ and by the line with equation $x = a$

The curves C_1 and C_2 intersect at the point B, with x coordinate p, where $p < a$

(c) Find the value of p.

(2)

The region R is rotated through 360° about the x-axis to generate a solid with volume $\frac{27\pi}{64}$

(d) Use algebraic integration to find the value of a.

(7)

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14. Question 10 · part b · 4 marks · Working and answer

Working and answer

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15. Question 4 · part c · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

4 $f(x) = 2x^3 + ax^2 + bx + 18$ where a and b are constants

When $f'(x)$ is divided by $(x - 2)$ the remainder is 5

Given that $(x - 2)$ is a factor of $f(x)$

(a) find the value of a and the value of b.

(6)

(b) Express $f(x)$ as a product of linear factors.

(3)

(c) Hence use algebra to solve the equation $f(x) = 0$

(2)

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15. Question 4 · part c · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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15. Question 4 · part c · 2 marks · Working and answer

Working and answer

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16. Question 4 · part c · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

4 $f(x) = px^3 + qx^2 - 37x - 12q$ where p and q are constants.

When $f'(x)$ is divided by $(x + 2)$ the remainder is -33

Given that $(x + 5)$ is a factor of $f(x)$

(a) (i) show that $p = 2$

(ii) find the value of q

(6)

(b) Hence, use algebra to factorise $f(x)$ completely.

(3)

(c) Hence solve the equation $f(x) = 0$

(2)

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16. Question 4 · part c · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

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Question 4 continued

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(Total for Question 4 is 11 marks)

16. Question 4 · part c · 2 marks · Working and answer

Working and answer

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17. Question 7 · part b · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

7 $f(x) = 2x^3 + Ax^2 + Bx - 20$ where A and B are constants.

Given that

- $(2x+1)$ is a factor of $f(x)$
- when $f'(x)$ is divided by $(x-1)$ the remainder is -27

(a) show that $A = 3$ and find the value of B

(5)

(b) Hence, using algebra, solve the equation $f(x) = 0$

(3)

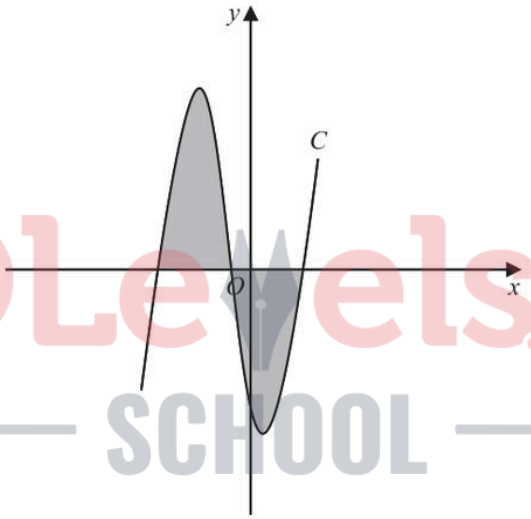


Figure 2

Figure 2 shows a sketch of part of the curve C with equation $y = f(x)$

(c) Use calculus to find the exact area of the finite region bounded by C and the x-axis, shown shaded in Figure 2

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17. Question 7 · part b · 3 marks · Working and answer

Working and answer

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18. Question 11 · part b · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

11 $f'(x) = ax^2 - 14x - 10$ where $a \in \mathbb{Z}$

Given that $(x - 4)$ is a factor of $f(x)$ and that when $f(x)$ is divided by $(x + 1)$ the remainder is 25

(a) show that $a = 6$

(6)

(b) Hence use algebra to solve the equation $f(x) = 0$

(6)

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18. Question 11 · part b · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 11 continued

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18. Question 11 · part b · 6 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 12 marks)

18. Question 11 · part b · 6 marks · Working and answer

Working and answer

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19. Question 7 · part b · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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19. Question 7 · part b · 4 marks · Working and answer

Working and answer

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20. Question 3 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

3 $g'(x) = mx^2 - 10x - 37$ where m is an integer

The curve $y = g(x)$ passes through the point with coordinates $(1, 20)$

Given that $(x - 5)$ is a factor of $g(x)$

(a) show that $g(x) = 2x^3 - 5x^2 - 37x + 60$

(5)

(b) Hence, or otherwise, use algebra to solve the equation $g(x) = 0$

(3)

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20. Question 3 · part b · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 8 marks)

20. Question 3 · part b · 3 marks · Working and answer

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21. Question 8 · part b · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

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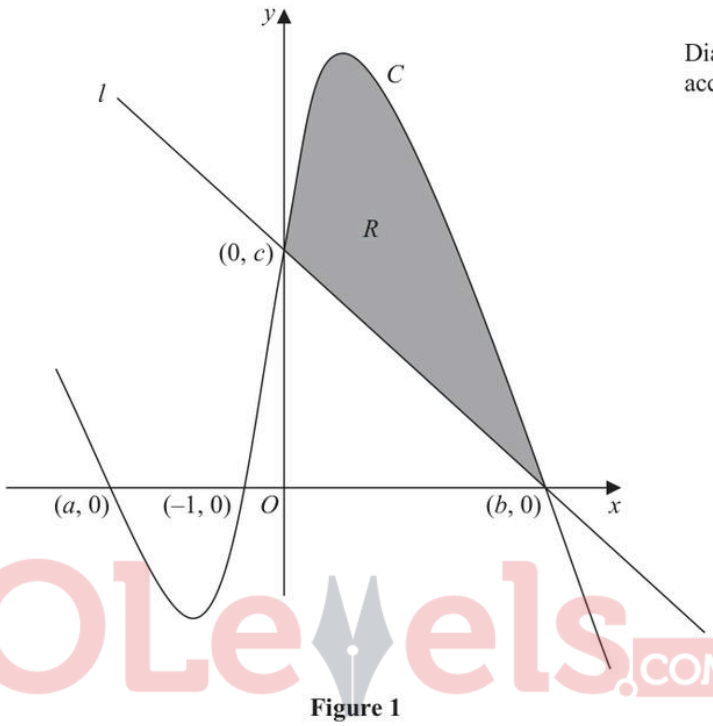


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve C with equation $y = f(x)$

The curve C passes through the points with coordinates

$$(a, 0), (-1, 0), (b, 0) \text{ and } (0, c)$$

Given that $f'(x) = 17 + 2x - 3x^2$

(a) show that the equation of C is $y = 15 + 17x + x^2 - x^3$

(4)

(b) Find the value of a, the value of b and the value of c

(6)

The straight line l intersects C at the points with coordinates $(b, 0)$ and $(0, c)$

The region R, shown shaded in Figure 1, is bounded by l and C

(c) Use algebraic integration to find the exact area of region R

(5)

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21. Question 8 · part b · 6 marks · Working and answer

Working and answer

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22. Question 9 · part c · 5 marks

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Further practice · Attempt part c only.

9 $f(x) = 3x^4 + 4x^3 - 36x^2 + 64$

Given that $f(x)$ can be written in the form $(x-2)^2(ax^2 + bx + c)$

(a) find the value of a, the value of b and the value of c

(4)

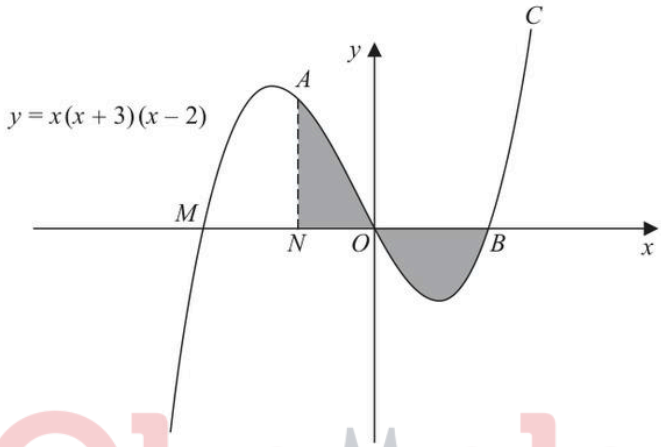


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a sketch of part of the curve C with equation $y = x(x+3)(x-2)$

The curve C crosses the x-axis at the point M, the origin and the point B.
The point N lies on the x-axis between M and O.
The point A lies on C such that AN is parallel to the y-axis.

The area of the shaded region bounded by the curve and OB is numerically equal to the area of the shaded region bounded by the curve, ON and NA.

Given that the coordinates of N are $(n, 0)$,

(b) use algebraic integration to show that n satisfies the equation

$$(x-2)^2(3x^2 + 16x + 16) = 0$$

(7)

(c) Hence find the exact coordinates of A.

(5)

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P 7 1 6 6 1 A 0 2 6 3 2

22. Question 9 · part c · 5 marks

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Further practice · Attempt part c only.

Question 9 continued

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22. Question 9 · part c · 5 marks

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Question 9 continued

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22. Question 9 · part c · 5 marks

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Question 9 continued

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(Total for Question 9 is 16 marks)

22. Question 9 · part c · 5 marks · Working and answer

Working and answer

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